

Hospital Patient Flow & Resource Analysis Report

Introduction

This project analysed patient admission and operational data from Lagos General Hospital covering the period from 2021 to 2023. The objective was to evaluate patient flow, understand resource utilization, identify revenue collection gaps, and provide insights that can support operational efficiency and healthcare decision-making. The analysis was performed using Python, MySQL, Excel, and Power BI.

Data Preparation

The dataset consisted of 60,000 patient records containing demographic information, diagnoses, admission details, outcomes, billing information, and readmission indicators. Data quality assessment showed no duplicate records. A small number of missing values were identified in the insurance type field and were handled appropriately during analysis. Date fields were converted to proper date formats, and a new variable called bill gap was created to measure the difference between billed amounts and actual payments received.

Key Findings

Monthly patient admissions remained relatively stable across the analysis period, suggesting consistent demand for healthcare services. A noticeable decline in January 2024 was observed, which likely reflects incomplete records outside the primary study period.

Most patients were successfully discharged, while deceased and discharge-against-medical-advice cases accounted for a much smaller proportion of outcomes.

Length of stay varied across departments. Emergency patients generally experienced shorter stays, while other departments showed wider variation in patient duration.

Analysis of insurance types revealed a balanced distribution of patients across payment categories. However, average billed amounts consistently exceeded average payments received, indicating the presence of revenue collection gaps.

Readmission patterns varied among departments, highlighting opportunities for targeted interventions and improved follow-up care.

Recommendations

1. Strengthen revenue collection processes to reduce outstanding balances and improve financial performance.
2. Monitor departments with higher readmission rates and implement targeted patient follow-up strategies.

3. Continue analysing patient flow patterns to support staffing and resource allocation decisions.
4. Investigate departments with longer average lengths of stay to identify opportunities for operational improvements.
5. Maintain diverse insurance partnerships to support broad patient access to healthcare services.

Conclusion

The analysis provided valuable insights into patient admissions, financial performance, and healthcare operations. The interactive dashboard and supporting analyses provide hospital management with data-driven tools for improving service delivery, optimizing resources, and strengthening revenue collection processes.